

Optiver Carbon Portfolio - Executive Summary

Challenge Objective

Build a portfolio delivering 100,000 tonnes CO₂e on a \$1,000,000 budget with 95% reliability under project failures.

Key Results

Metric	Value
Total Cost	\$181,658.56
Projects	13 credits
Nominal Tonnes	166,671 tCO ₂ e
Modelled Success Rate	98.3% (p=0.3)
95% CI Lower Bound	98.0%
5th Percentile Delivery	110,970 tonnes

Status: ✓ PASS

Portfolio Quality Signals

- Tonnes-weighted metadata score: 0.42/1.0
- Metadata heuristic, not a certified quality rating; unused by the optimiser.
- Removal Projects: 0.0% of portfolio
- Known-vintage weighted average: 2011

Top Holdings (by cost)

Project	Country	Tonnes	Cost
Xundian Jinfeng 12.6MW Hydropower Project	China	16,667	\$20,000
Canakkale WPP	Türkiye	16,667	\$19,834
Inner Mongolia Sunjiaying 50.25MW Wind Power Project	China	16,667	\$18,334
Monjolinho Energética S/A Hydropower Plant Project (Alzir dos Santos Antunes)	Brazil	16,295	\$17,599
2x50 MW Orange Suvaan Solar Photovoltaic Power Project in Maharashtra India	India	16,667	\$17,500

Risk Management

- Diversification: Spread across countries, developers, registries, and project types
- Buffer Pools: 50% recovery on projects with buffer protection
- Stress Testing: Validated under multiple correlation scenarios

Method Summary

1. Baselines: Compare cheapest nominal vs. expected delivery approaches
2. Search: Bounded heuristic across 6 allocation templates
3. Evaluation: 10,000 Monte Carlo scenarios per correlation setting
4. Validation: Wilson score 95% confidence intervals

Generated for AdaHack 2026 | Optiver Challenge | Modelled results, not real-world guarantees